

BranDive Media Solutions

Weekly Development Sprint – Project 1

CampFlow™

Campground Booking & Reservation Platform

Project Duration: 7 Days

Difficulty: Intermediate

Project Type: Real Client Case Study

Role: Full Stack Developer

Project Overview

Congratulations! Your team has been assigned your first client project.

You are working as a Full Stack Developer at **BranDive Media Solutions**.

A client has approached BranDive requesting a complete digital solution for their campground business. Your responsibility is to analyze the client's requirements, design the system, develop the required modules, and present a professional solution within one week.

This project simulates a real-world client engagement. Treat every requirement as if you are delivering a production-ready system to a paying client.

Client Background

Client Name: Green Valley Campgrounds

Business: Campground & Outdoor Resort

Locations: 4 Campgrounds

Country: United States

Green Valley Campgrounds has been operating successfully for over 12 years. The business offers RV camping, tent sites, cabins, glamping tents, and outdoor activities. Due to increasing demand, the owner has expanded to four different locations across the state.

Currently, reservations are handled manually through phone calls, Facebook messages, and email, resulting in scheduling conflicts and a poor customer experience.

The client wants a modern web platform that allows guests to easily explore campgrounds, check availability, make reservations, and complete payments online.

Current Business Challenges

The client currently faces the following issues:

- Reservations are managed manually.
 - Customers cannot view campsite availability online.
 - Payments are collected manually.
 - Staff spend hours answering repetitive questions.
 - There is no centralized reservation system.
 - Managing four different campground locations is difficult.
 - Seasonal pricing is maintained manually in spreadsheets.
 - Owners have no reporting or analytics.
 - There is no customer portal for previous bookings.
 - The current website is outdated and not mobile-friendly.
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Project Objectives

Your goal is to develop a centralized campground reservation platform that enables customers to book campsites online while allowing administrators to efficiently manage locations, reservations, pricing, and customer information.

The system should improve operational efficiency, enhance customer experience, and provide the business with valuable operational insights.

Project Scope

Your solution should include:

- Public Website
- Reservation System

- Customer Portal
 - Admin Dashboard
 - Campground Management
 - Seasonal Pricing Management
 - Interactive Campsite Map
 - Payment Integration (UI Only)
 - Analytics Dashboard
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User Roles

Your system should support the following users.

Guest Visitor

Can:

- Browse campgrounds
 - Search availability
 - View campsite details
 - Register
 - Book campsites
 - Make payments
 - Contact support
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Registered Customer

Can:

- Login
 - Manage profile
 - View booking history
 - Download invoices
 - Cancel reservations
 - Leave reviews
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Campground Manager

Can:

- Manage reservations
- Update campsite availability

- Manage pricing
 - View customer information
 - Approve or reject bookings
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Administrator

Can:

- Manage all locations
 - Manage users
 - View reports
 - Configure seasonal pricing
 - Create promotional offers
 - View revenue analytics
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Functional Requirements

Module 1 – Public Website

Create a modern responsive website containing:

- Homepage
 - About Us
 - Campground Locations
 - Campsite Categories
 - Amenities
 - Activities
 - Pricing
 - Gallery
 - FAQ
 - Contact Page
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Module 2 – Reservation System

The reservation system should allow users to:

- Select campground
- Choose campsite
- Select arrival date
- Select departure date
- Select number of guests

- Check availability
 - Confirm reservation
 - Receive confirmation
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Module 3 – Interactive Camp Map

Design an interactive map where users can:

- View campsite locations
 - View occupied sites
 - View available sites
 - Select preferred campsite
 - Filter by campsite type
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Module 4 – Customer Dashboard

Include:

- Upcoming Reservations
 - Booking History
 - Profile Settings
 - Saved Payment Methods
 - Favorite Campsites
 - Reviews
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Module 5 – Admin Dashboard

The administrator dashboard should include:

- Total Reservations
 - Revenue Overview
 - Occupancy Rate
 - Customer Statistics
 - Campground Performance
 - Seasonal Reports
 - Booking Calendar
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Module 6 – Pricing Management

Allow administrators to:

- Create seasons
 - Apply seasonal pricing
 - Weekend pricing
 - Holiday pricing
 - Promotional discounts
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Module 7 – Notification System

The system should send:

- Booking Confirmation
- Cancellation Confirmation
- Reminder Emails
- Payment Confirmation

(UI Design Only)

Non-Functional Requirements

The application should be:

- Mobile Responsive
 - Fast Loading
 - User Friendly
 - Secure
 - Scalable
 - Clean UI
 - Professional Design
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Suggested Pages

Minimum Pages:

- Home
- Campgrounds
- Campsite Details
- Reservation

- Login
 - Register
 - Customer Dashboard
 - Admin Dashboard
 - Pricing
 - Contact
 - About
 - FAQ
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Suggested Database Structure

Include tables for:

- Users
- Customers
- Campgrounds
- Campsites
- Reservations
- Payments
- Pricing Seasons
- Reviews
- Activities

Design an ER Diagram showing relationships between entities.

Suggested Technology Stack

Frontend

- React.js
- Next.js
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT

Version Control

- Git & GitHub

Hosting

- Vercel
 - Hostinger VPS
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UI/UX Expectations

Your interface should be modern and suitable for the tourism and hospitality industry.

Focus on:

- Excellent user experience
 - Easy navigation
 - Mobile-first design
 - Attractive visuals
 - Consistent color palette
 - Professional typography
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Deliverables

Each intern/team must submit:

- Complete Source Code
 - GitHub Repository
 - README Documentation
 - Database Schema
 - ER Diagram
 - User Flow Diagram
 - Wireframes (Optional)
 - Screenshots
 - Live Deployment (Bonus)
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7-Day Sprint Plan

Day 1 (15%)

Understand the requirements, research competitors, create sitemap, user flow, and low-fidelity wireframes.

Day 2 (15%)

Design all UI screens and build the frontend structure with responsive layouts.

Day 3 (15%)

Develop the public website, navigation, and static pages.

Day 4 (20%)

Implement the reservation flow, customer authentication, and dashboard interfaces.

Day 5 (15%)

Develop the admin dashboard, campsite management, and pricing management screens.

Day 6 (10%)

Implement remaining modules, refine the UI, fix bugs, and optimize responsiveness.

Day 7 (10%)

Testing, documentation, GitHub cleanup, deployment (if applicable), and final presentation.

Bonus Features

You may earn additional marks by implementing:

- Google Maps Integration
- Weather API
- Dark Mode
- Wishlist
- Coupon System
- Multi-language Support
- AI Chat Assistant
- Online Payments (Sandbox)
- Email Notifications

Evaluation Criteria

Criteria	Marks
Requirement Understanding	15
UI / UX Design	20
Frontend Development	20
Backend Structure	15
Database Design	10
Code Quality	10
Documentation	5
Presentation	5

Total: 100 Marks

Submission Requirements

Submit the following within **7 days**:

- GitHub Repository Link
- Live Deployment (Optional but Recommended)
- PDF Documentation
- ER Diagram
- Project Presentation (Maximum 10 Slides)

- Screenshots
- README File

Late submissions may result in a deduction of marks unless prior approval has been obtained from the Team Lead.

Client Acceptance Criteria

The client will consider the project successful if:

- Customers can browse and reserve campsites without confusion.
 - The reservation process is intuitive and responsive.
 - Administrators can manage all four campground locations from one dashboard.
 - Seasonal pricing can be configured easily.
 - The interface is modern, responsive, and professional.
 - The architecture is scalable enough to support future features such as mobile applications, AI assistants, loyalty programs, and online payments.
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Note from BranDive Media Solutions

This project is intended to simulate a real client engagement. Your solution should demonstrate not only technical ability but also your understanding of business requirements, user experience, scalability, and professional software development practices. Approach the project as if you are delivering it to a paying client, with attention to quality, usability, and documentation.